HAPPIEST HEALTH · DIGITAL TWIN CHALLENGE 2026

# GlucoTwin

A continuously updated virtual Type 2 diabetes patient that estimates, up to 2 hours ahead, the risk that a post-meal glucose excursion is about to begin, and shows a doctor why.

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**Registered team:** Mohnish Shanmugavadivel (Team Leader, primary/sole developer) · Sachithanantham V (Registered Team Member)

Implementation and experiments: Mohnish Shanmugavadivel

Research and educational prototype on **synthetic patients only**. Not a medical device. Outputs are not diagnoses or treatment instructions.

# Problem and healthcare use case

## Condition: Type 2 diabetes (India)

- Post-meal glucose excursions are common in T2D, and a clinician reviewing CGM usually sees them **after** they happen.
- The challenge asks for a twin that fuses **simulated EHR** with **wearable / CGM streams** and predicts an adverse event, e.g. "a sudden glucose spike 2 hours in advance".

## Use case

- A doctor monitors several remotely monitored patients.
- The twin flags who is likely to cross 180 mg/dL in the next 2 h, while glucose is still normal, with a forecast band and the reasons.

## Event definition (frozen before modelling)

At time t, asked only if the latest CGM ≤ 180 mg/dL. **Positive** if ≥ 3 consecutive 5-min CGM readings > 180 mg/dL (≥ 15 min; gaps break the run) **begin** within (t, t + 120 min].

Why "onset": predicting "above 180" while already above 180 is trivial and useless clinically.

## Outputs per CGM reading

- Glucose forecast at 30/60/90/120 min with an 80% interval
- Probability of excursion onset within 2 h, alert at a validation-chosen threshold
- Grouped explanation (associations, not causes)

# Why a digital twin, and not a single risk score?

## Continuously updated

A versioned per-patient state is updated with **every** observation: glucose and trend, 24 h time-in-range, meals and carbs on board, activity, data quality. Forecasts are re-issued at each reading.

## Fuses record + streams

The static record (labs, medications, history) gives context and a prior. Streams give the current state. Both feed the same leakage-safe feature builder offline and online.

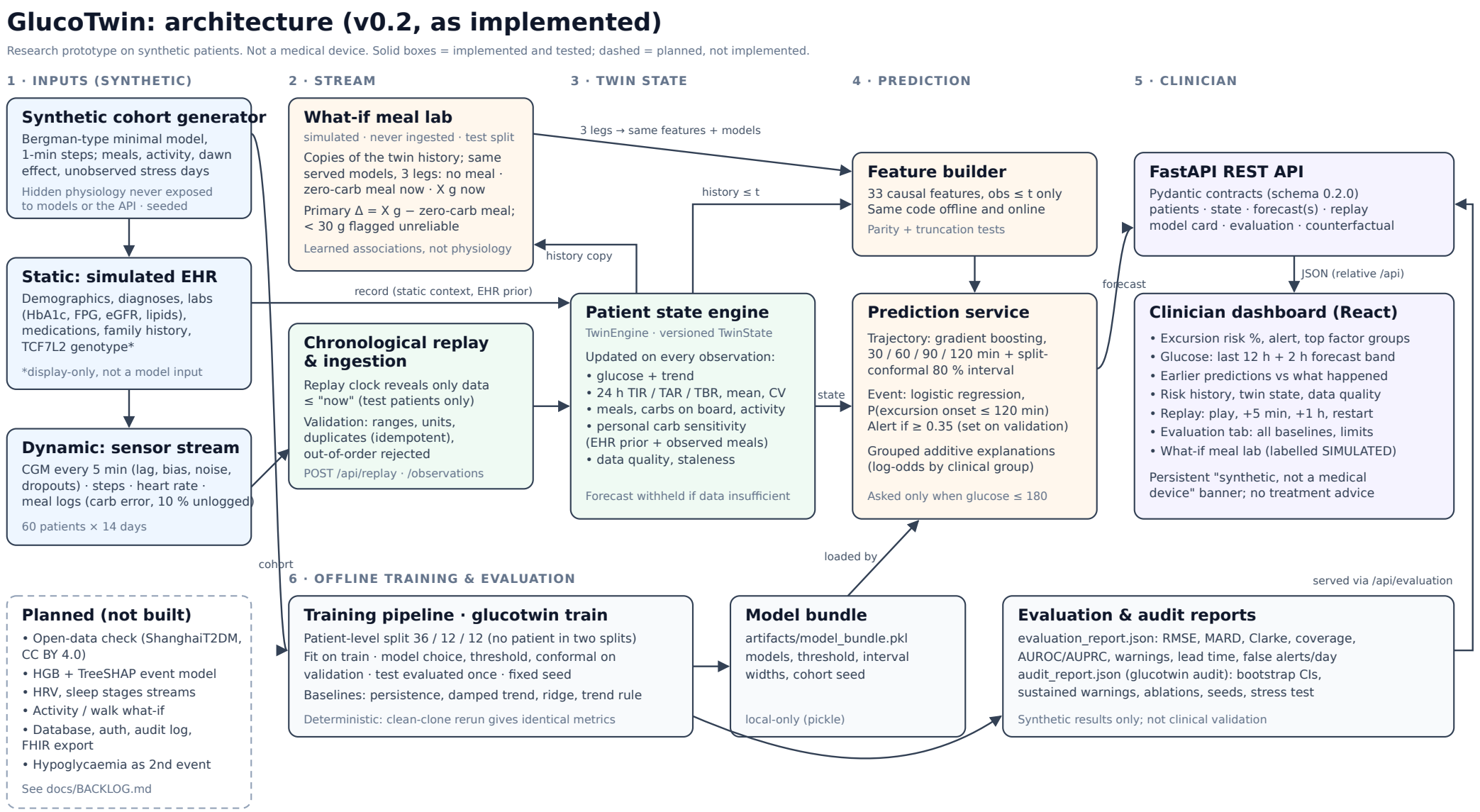
## Personalised

Each twin learns its patient's **carb sensitivity** (mg/dL per g) from observed meal responses, shrunk towards a record-based prior. This changes the expected post-meal rise.

## What it is not (yet)

The twin does not simulate physiological interventions such as activity ("what if she walks after lunch?"). That would need per-patient physiological parameter fitting, which is on the roadmap, not built. Its what-if meal lab only re-runs the learned forecasters on a hypothetical meal (an association, not a mechanism). Our twin is a **state-tracking forecaster** with one personalised physiological parameter.

# Architecture (as implemented)



# Data: synthetic, by rule and by necessity

- Rules: "strictly required to use anonymized, open-source, or synthetic datasets".
- Real CGM + EHR datasets with 5-min data (OhioT1DM, T1DEXI, DiaTrend) need data use agreements, which is not feasible before the deadline.
- **Generator:** a Bergman-type minimal model with meals, activity, dawn effect and unobserved "stress days". **Hidden** physiology is never shown to models.
- **Observable:** record (labs derived with noise), CGM with lag, bias, AR noise and dropouts, steps, heart rate, meal logs with carb error and 10% unlogged.
- 60 patients × 14 days; 5-min CGM; seed 20261017.

| Plausibility check | Ours | ShanghaiT2DM* |
|---|---|---|
| Time in range 70–180 | 77.8% | 77.7% |
| Time above 180 | 21.8% | 20.0% |
| Time below 70 | 0.4% | 2.4% |

*Published summary of an open Chinese T2D cohort (Zhao et al., Sci Data 2023), quoted from a secondary copy. This is a plausibility check, not validation.

**Known gaps.** Lows are under-represented: hypoglycaemia only occurs on sulfonylurea or insulin in our simulator. Synthetic HbA1c is derived from the evaluated 14 days (corr 0.95 with that window's mean CGM). The ablation shows it does not inflate results (slide 8).

# Evaluation protocol and leakage safeguards

## Protocol

- **Patient-level split** 36 / 12 / 12; no patient appears in two splits.
- All models are fitted on train only. **Model choice, alert threshold and conformal intervals use validation only.** Test is evaluated once.
- Baselines on the identical test rows: persistence, damped trend, ridge (trajectory); trend rule (event).
- Deterministic: a **clean clone** with a fresh environment reproduced every metric exactly (wave-2 audit).

## Leakage safeguards (tested)

- Features use observations ≤ t only. A **truncation test** shows that removing the future never changes past features.
- The **same code** builds features offline and online (parity test).
- Personal carb sensitivity uses a meal only after its 2-h response window has ended.
- **Mutation check:** injecting a 30-min future shift makes two leakage tests fail.
- The dashboard replay can only show held-out test patients and never reveals data after "now" (API test).

# Results on 12 held-out synthetic patients

reproduced in a clean clone

## Glucose forecast RMSE (mg/dL)

| Model | 30 min | 60 min | 120 min |
|---|---|---|---|
| Persistence (last value) | 18.7 | 31.4 | 46.6 |
| Damped trend | 17.4 | 31.1 | 50.4 |
| Ridge | 11.2 | 17.0 | 27.9 |
| **Gradient boosting (served)** | **10.4** | **17.2** | **25.9** |

95% patient-bootstrap CI, served model: 30 min 9.8–11.0; 120 min 23.2–29.1. 80% conformal interval coverage on test: 0.76–0.80. Clarke A+B ≥ 95.7%.

## Excursion onset within 120 min

| Model | AUROC (95% CI) | Sens / Spec | False alerts / day |
|---|---|---|---|
| Trend rule baseline | 0.655 (0.583–0.736) | 0.44 / 0.74 | 6.3 |
| **Logistic regression (served)** | **0.884 (0.839–0.923)** | **0.67 / 0.90** | **1.4** |
| Gradient boosting | 0.936 (0.915–0.955) | 0.78 / 0.90 | 1.1 |

Test prevalence 14.6% of eligible time points. Logistic regression is served because its explanation is exact and additive. Gradient boosting discriminates better (roadmap: serve it with SHAP).

These describe our simulator and are an **upper bound**. Real-data studies report a 30-min RMSE of about 17–20 mg/dL. Not clinical validation.

Source: reports/evaluation_report.json, reports/audit_report.json

# A stricter look: our own audit

wave-2 · glucotwin audit

## How often does the doctor get warned? (served model)

| Definition | Share of onsets |
|---|---|
| Any alert in the 2 h before onset | 91% (0.83–0.98) |
| …with ≥ 30 min lead | 87% |
| **Sustained: alert still on just before onset** | **69% (0.48–0.83)** |

Median lead: any alert 95 min; sustained 85 min (capped at 120). False-alert episodes per monitored day: 1.36 (0.92–1.92). The headline "91% warned" uses the lenient definition, so we report both.

## Robustness

| Check | Logistic AUROC | Sustained warned |
|---|---|---|
| Main cohort | 0.884 | 69% |
| 3 other cohort seeds | 0.883–0.910 | 72%–78% |
| No record (EHR) features | 0.907 | 73% |
| No meal features | 0.849 | 68% |
| Stress: 60% logged, ±50% carbs, 0–45 min late, noisier CGM | 0.904 | 70% |

Stress test: RMSE at 30 min rises from 10.4 to 15.4; warnings with ≥ 30 min lead fall from 87% to 78%. Direct record features did **not** help held-out discrimination; the record contributes through the personal prior and context.

Source: reports/audit_report.json (deterministic; 1000 patient-level bootstrap resamples)

# Twin validity against simulator ground truth

synthetic only · 95% patient-bootstrap CIs

## Does it learn the person?

- Hidden per-patient meal gain recovered: r = 0.62 (record prior) → 0.95 after 3 days → 0.98 at 14 days.
- **After meals it matters:** switching personalisation off raises +60-min RMSE by **+2.44 mg/dL** (0.73–3.86; 20.52 → 22.96), in 6,266 test moments with ≥ 10 g logged in the past hour (post-hoc).
- Confidence shown on screen, e.g. SYN-0039: **87%** from 20 own meals, ± 0.16 sampling error (not a probability).

## Is the What-If right?

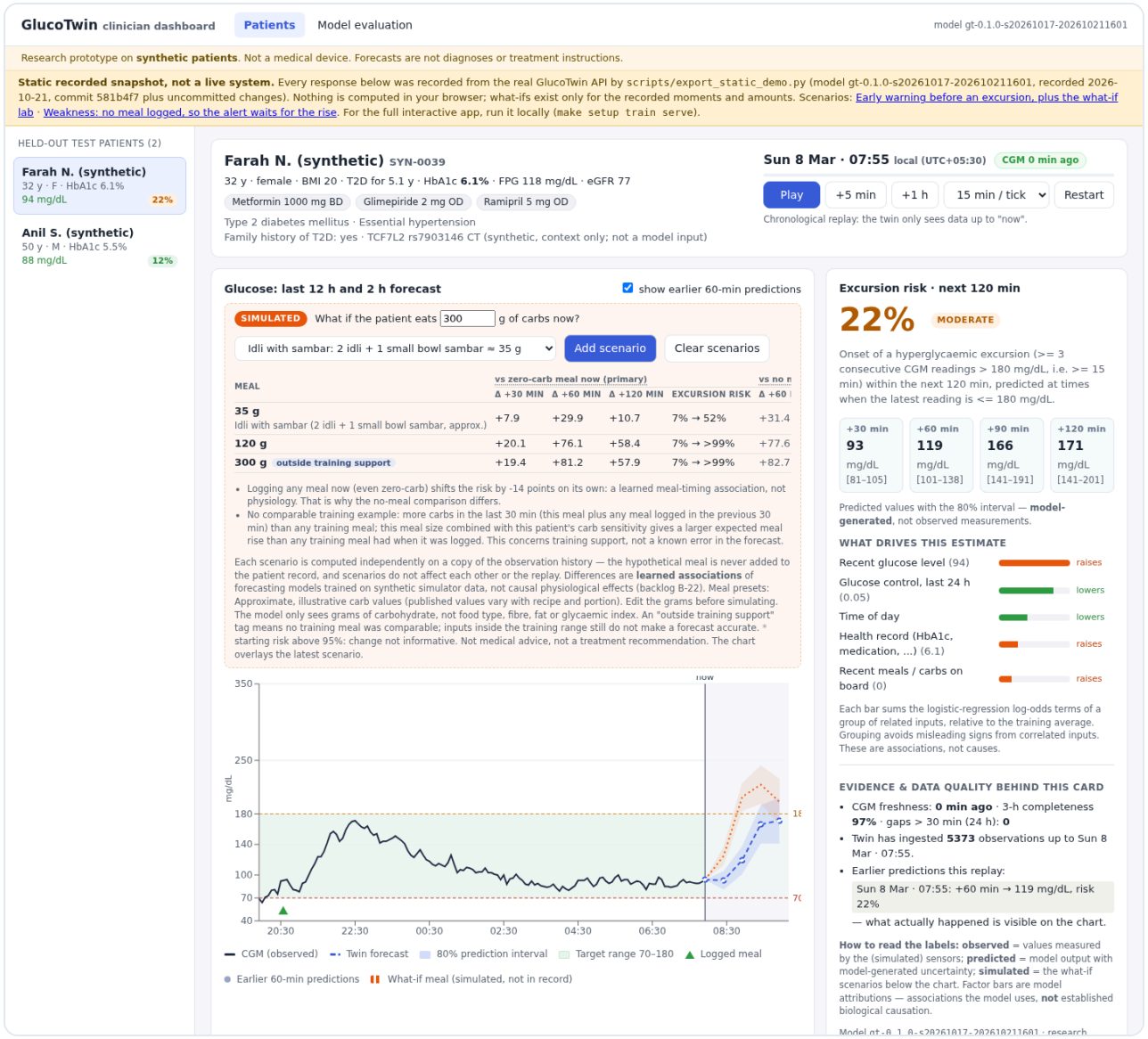
- **Served What-If (negative result):** at +120 min, MAE 45.3 vs 20.8 mg/dL for a cohort-average curve (difference +24.5 [+13.1, +35.4]; 20–120 g).
- **Train-fitted personalised curve** (shape from 36 training patients' observed CGM; variant chosen on 12 validation patients: median shape, validation-calibrated; scored once on 12 test patients): +120 min MAE **5.9** vs cohort 16.8 (difference −10.9 [−15.8, −6.2]) and served What-If 37.3.
- +30 / +60 min: curve 1.8 / 5.3 vs cohort 3.1 / 12.5. +30 min CI includes zero. Research column, not served; the simulator's near-linear meal response favours it.

## Interoperability and safety signals

- **FHIR R4 record** (one click, Copy/Download): 2 Condition · 3 MedicationStatement · 75 Observation · 1 Patient · 1 RiskAssessment. RiskAssessment carries the excursion probability (probabilityDecimal 0.215 at the replay start). Own synthetic ID, never ABHA; not ABDM-certified.
- **Low-glucose watch**: flags when the 80% lower bound falls below 70 mg/dL (risk card and each What-If row), with sulfonylurea/insulin context. Not a probability: on test patients a low followed in 33.5% of flagged moments.

Sources: reports/research/twin_validity.json, twin_and_whatif_validation.json, frontend/static-demo/demo/*.json

# Clinician dashboard: what the doctor sees




- **Patient list:** held-out test patients, latest glucose, latest risk.
- **Record header:** labs, medications, history.
- **Glucose chart:** last 12 h, 2-h forecast with 80% band, meals, earlier predictions versus what happened.
- **Risk card:** probability (never shown as 100%), alert wording "Consider reviewing", top factor groups.
- **Risk history** and **twin state:** TIR, carbs on board, learned carb sensitivity, data quality.
- **Replay:** the twin only sees data up to "now".
- **Evaluation tab:** every baseline and limitation.
- **What-if meal lab** (simulated): *X* g eaten now versus a zero-carb meal at the same time, plus a grey no-meal comparison. Learned model associations, not physiology; meals under 30 g flagged unreliable.

# Twin-curve What-If, Indian meals, AGP and a fact-grounded handover

synthetic data · research estimates labelled

SIMULATED What if the patient eats 68 g of carbs now?

Rajma chawal: 1 plate ≈ 68 g · Mixed GI · team est.

Mixed GI · team estimate | compare with a post-meal walk (+20 min, 2,500 steps) | Add scenario

Clear scenarios

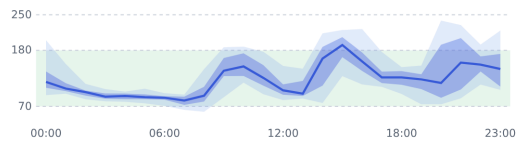
← Swipe the table for more columns (no-meal comparison, low watch) →

| MEAL | vs zero-carb meal now (primary) | | | | Personalised twin curve (research) | | |
|---|---|---|---|---|---|---|---|
| | Δ +30 MIN | Δ +60 MIN | Δ +120 MIN | EXCURSION RISK | Δ +60 MIN | Δ +120 MIN | + WALK, Δ +120 M |
| **35 g** Idli with sambar (2 idli + 1 small bowl sambar, approx.) | +7.9 | +29.9 | +10.7 | 7% → 52% | +42.5 | +60.0 | +48.8 (walk -11.2 |
| **68 g** Rajma chawal (1 plate, approx.) | +12.4 | +64.4 | +38.7 | 7% → 93% | +82.6 | +116.5 | +105.3 (walk -11. |

**Glucose profile (AGP, up to 14 days)**

as of Sun 8 Mar · 07:55 · recorded at the replay start

| TIME IN RANGE 70–180 | ABOVE 180 / 250 | BELOW 70 / 54 |
|---|---|---|
| 88.2% | 10.4% / 0% | 1.3% / 0.1% |
| **MEAN · CV** | **GMI (ESTIMATE)** | **DATA** |
| 122.3 mg/dL · 31.9% | 6.2% | 6.3 d · 44.9% coverage |




Bands: 5–95th and 25–75th percentiles, line: median, by hour of the local day. Metrics follow the 2019 international consensus on CGM reporting; **this window is below its 14-day / 70% data minimum, so read it as indicative only.** Synthetic CGM; descriptive, not used by any forecast, not a diagnosis.

**▾ Clinician handover note** (fact-grounded, fixed template)

✓ Verified: 100% of numbers (49/49) trace to the twin record, state or forecast (43) or a stated definition (6).

- Handover: Farah N. (synthetic), SYN-0039; 32 y; diabetes duration 5.1 y. Twin state at 07:55 local time (UTC+05:30).
- Latest CGM: 94 mg/dL, stable (+0.5 mg/dL/min). Last reading 0 min ago; CGM completeness over 3 h 97%.
- Past 24 h (285 readings): mean 116 mg/dL; time in 70-180 range 94%, above 5%, below 1%.
- Model forecast (synthetic-trained, not validated on real patients): +30 min 93, +60 min 119, +120 min 171 mg/dL; 80% prediction interval at +120 min 141-201 mg/dL.
- Estimated risk of glucose above 180 mg/dL within 120 min: 22% (moderate band; alert threshold 35%; no alert).
- Low watch: the lower edge of the prediction interval stays at or above 70 mg/dL (not a guarantee).
- Context: carbs on board 0 g; last logged meal 681 min ago; 655 steps in the past hour; personal carb sensitivity 1.70 mg/dL per g from 20 meals.
- Glucose profile (14-day window, as of 07:55): 6.3 days of data, coverage 44.9% (below the 14-day / 70% consensus minimum); time in range 88.2%; GMI 6.2% (estimate from mean glucose, not a laboratory result).
- Medications on record: Metformin 1000 mg BD; Glimepiride 2 mg OD; Ramipril 5 mg OD.
- Generated by a fixed template from the twin's own record, state and forecast; synthetic data; research prototype. Describes model output only; not a

- **Twin curve beside the served What-If.** SYN-0039, 35 g at 07:55, +120 min: served +10.7, twin curve +60.0, paired-simulator truth +65.4. Test MAE 5.9 vs 37.3.
- **Post-meal walk toggle** (20 min, 2,500 steps): cohort-level effect; test MAE 4.3 vs 11.6 for "no effect" at +120 min. Not fed to the tree models.
- **24 regional Indian dishes** (South 10 · North 5 · West 5 · East 2 · Snacks 2): 11 cited, 13 team estimates; indicative GI badges. A swap finder ranks lower-carb presets by the twin curve (in effect, by grams).
- **AGP card:** time in ranges, GMI, CV, hourly percentiles; flags data below the 14-day / 70% minimum.
- **Clinician + Atithi (patient) notes:** fixed templates; one parser traces every number to the twin, else the note is withheld. Describe, never prescribe.
- **FHIR in:** `POST /api/fhir/Bundle` validates R4 Observations (CGM 99504-3; 14749-6 with a lab-code warning; HR 8867-4; HbA1c 4548-4). Nothing stored; not certified.

Sources: frontend/static-demo/demo/early-warning.json, reports/research/walk_whatif_validation.json, docs/research/TWIN_VALIDITY.md (addenda v3, v4)

# Two demo moments: an early warning and a late one

### SYN-0039 · Sun 8 Mar (held-out patient)

- 08:53 a 140 g breakfast is logged; at 08:55 glucose is **89 mg/dL**.
- Risk is **>99%**, the alert switches on, and the forecast reaches about 205 mg/dL at +60 min.
- Onset at 10:20, peak about 207 mg/dL. The alert stayed on throughout: an **85-min sustained warning**.
- Honest detail: the rise came later than the 30-min forecast (103 observed versus 128 forecast at 09:25).

Deep link /?patient=SYN-0039&start=151.833 · verified against the live API

### SYN-0019 · Thu 12 Mar (held-out patient)

- 14:45: glucose 88 mg/dL, risk 31%, just **below** the 35% threshold. **No meal logged.**
- 14:50: glucose rising fast, risk 36%: the alert fires, only 45 min before the first reading above 180 (15:35); peak about 191. **Warned, but late.**
- Unlogged meals are the main failure mode: removing meal features costs the most AUROC, and the stress test cuts lead time.

Deep link /?patient=SYN-0019&start=253.667 · Of 298 test onsets, 205 (69%) had a sustained warning and 272 had an alert at some point in the prior 2 h. scripts/find_demo_scenario.py lists 53 daytime onsets (08:00–21:59, ≥ 30 h into the data) whose alert was off at the last reading before onset.

# Safety boundaries and limitations

## Safety boundaries (built in)

- A "synthetic, not a medical device" banner on every screen; synthetic IDs (SYN-####) and names tagged " (synthetic)".
- No instructions: the alert text is "Consider reviewing", never "give insulin" or "eat".
- Explanations and what-if meal results are labelled as associations, not causes; a what-if is never written to the record.
- No risk number when data are insufficient or stale; no onset risk when glucose is already above 180.
- Probabilities are never displayed as 100% or 0%.
- Local demo: no authentication, so it is **not for real patient data**.

## Limitations

- Evaluated only on our own simulator; no real-patient or external validation.
- 12 test patients, so the confidence intervals are wide.
- Meal logging is near-real-time in the main cohort, which is optimistic.
- Lows are under-represented (0.37% of readings < 70 mg/dL vs 2.4% published); the low-glucose watch is a lower bound, **not a probability**.
- Vs simulator ground truth, the What-If underestimates late meal effects (+120 min MAE 29.0 vs 12.8 mg/dL for a cohort-average curve).
- Personalisation is shown only in simulation (hidden meal gain recovered, r 0.95 by day 3; +30 min RMSE −0.52 mg/dL, 95% CI 0.12–0.96).
- No HRV or sleep streams; genotype is synthetic and display-only. No regulatory claims (CDSCO SaMD, DPDP, HIPAA).

# Potential impact and the path to real validation

## If validated, the intended value is

- Earlier, reviewable warnings for remotely monitored T2D patients, in place of after-the-fact CGM review.
- A per-patient view that explains *why* risk is high (meal, trend, recent control).
- A transparent evaluation template (baselines, CIs, alert burden) for any future model.

We have run no pilot and have no user or outcome data. No impact figures are claimed.

## Validation roadmap (planned)

1. Retrospective check on open data (ShanghaiT2DM, CC BY 4.0; 15-min CGM).
2. Serve gradient boosting with SHAP; within-patient chronological evaluation.
3. Clinician usability review of the dashboard (think-aloud, no patients).
4. Ethics-approved retrospective study on consented, de-identified data.
5. Prospective **silent-mode** study (alerts not shown to clinicians), then a regulatory pathway assessment.

# Engineering and reproducibility

## Stack

- Python 3.11 · NumPy · pandas · scikit-learn · Pydantic v2 contracts · FastAPI
- React 19 + TypeScript + Recharts (Vite)
- pytest (161 tests) · 49 frontend tests · ruff · GitHub Actions CI · pinned requirements.lock / package-lock.json

## Reproduce

```
make setup
make train     # same metrics every run
make serve     # http://localhost:8000
glucotwin audit  # CIs, ablations, stress test
make check     # lint + tests + build
```

## Verified in the wave-2 audit

- Clean clone + fresh environment: tests pass and the report is identical.
- Audit re-implementation matches every headline metric exactly.
- pip-audit and npm audit: no known vulnerabilities (after a pytest upgrade).
- Fixed: API returned 500 on extreme replay offsets (now 422); training patients could be replayed (now refused).
- Open source: MIT licence.

# Thank you

GlucoTwin: a transparent, reproducible digital-twin prototype for early warning of post-meal hyperglycaemia.

**Team:** team mono · Mohnish Shanmugavadivel (Team Leader, primary/sole developer) · Sachithanantham V (Registered Team Member) — Amrita Vishwa Vidyapeetham, Chennai

All results are from synthetic patients. Not a medical device. Not clinical validation.